

LAEEQ AHMED

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PROFESSIONAL SUMMARY

Results-driven **AI/ML Engineer** with hands-on experience building autonomous AI agents, NLP pipelines, and real-time computer vision systems. Proficient in **Python**, **PyTorch**, and orchestrating complex LLM workflows using **LangGraph** and **LangChain**. Expertise in **Retrieval-Augmented Generation (RAG)**, and managing **Vector Databases**. Adept at bridging the gap between machine learning and production by deploying highly available AI microservices and inference APIs using **FastAPI**, **Docker**, and **AWS**.

EDUCATION

Matrusri Engineering College (Osmania University)

Hyderabad, India

Bachelor of Engineering (B.E.) in Information Technology | CGPA: 8.23/10

Nov 2022 – Jun 2026

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, JavaScript

AI/ML & Frameworks: PyTorch, TensorFlow, LangGraph, LangChain, Hugging Face, OpenCV, YOLOv8

LLMs & Generative AI: RAG, NLP, Prompt Engineering, Ollama, Llama-3, API Integration

Backend & Databases: FastAPI, Vector Databases (ChromaDB), Node.js, PostgreSQL, MongoDB

MLOps & Cloud: AWS, Docker, Git, CI/CD (GitHub Actions), RESTful APIs, Model Deployment

EXPERIENCE

Swecha & IIIT Hyderabad (Viswam AI)

Hyderabad, India

AI Developer Intern

Jun 2025 – Jul 2025

- Collaborated with a cross-functional team in an Agile environment to procure, clean, and anonymize 5,000+ SQL queries, establishing a secure, high-quality dataset for local LLM fine-tuning.
- Engineered an autonomous Text-to-SQL agent workflow using **LangGraph** and **Python**, integrating a fine-tuned **Llama-3 (3B)** model via **QLoRA** to translate natural language into optimized database queries.
- Implemented structured evaluation workflows to achieve **92% execution accuracy**; built a **FastAPI** backend containerized with **Docker** to enable automated testing and scalable model inference.

PROJECTS

CareerX AI – LLM-Powered Career Intelligence Platform

Live Demo | GitHub

Python, RAG, ChromaDB, LangChain, LLM APIs (OpenRouter), Node.js, Docker

- Architected an AI-driven platform utilizing **Retrieval-Augmented Generation (RAG)** and multi-model LLM APIs to deliver **4 core generative modules**, processing context-aware mock interviews and ATS analysis in **under 3 seconds**.
- Built an advanced retrieval pipeline utilizing **ChromaDB** to store and query high-dimensional text embeddings, achieving vector search latency for highly accurate semantic similarity scoring.
- Engineered complex prompt templates and structured JSON parsers to process unstructured user inputs with **99% parsing reliability**, exposing AI capabilities through **15+ secure RESTful APIs**.
- Containerized the AI microservices and database instances using **Docker**, automating deployment workflows via **GitHub Actions** to accelerate release cycles by **30%** and maintain a reliable CI/CD pipeline.

SmartFirePredict – Real-Time Computer Vision & IoT System

Live Demo | GitHub

Python, PyTorch, YOLOv8, OpenCV, FastAPI, ESP32, AWS

- Developed a real-time Computer Vision inference pipeline from scratch utilizing **Python**, **YOLOv8**, and **OpenCV**, achieving an **87% mAP50** score for high-accuracy fire and smoke detection.
- Designed a multi-threaded **FastAPI** inference server capable of processing concurrent high-resolution RTSP and WebSocket video streams, optimizing logic to reduce end-to-end latency by **1-2 seconds**.
- Engineered a multimodal sensor fusion algorithm combining CV bounding box data with hardware telemetry from ESP32 microcontrollers to reduce false positives by **20-30%**.
- Packaged the deep learning models and API endpoints into **Docker** containers, deploying the highly available inference engine to **AWS EC2** to ensure low-latency edge-to-cloud communication.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Tech Tycoons Club, Matrusri Engineering College

Hyderabad

Tech Executive

2024 – 2026

- Organized campus-wide hackathons and coding contests, managing cross-functional student teams.
- Conducted hands-on workshops on **Python**, **Algorithmic Problem Solving**, and **Git/GitHub** workflows to support peer learning and technical development.

ACHIEVEMENTS

- Achieved **Global Rank 1130** in **TCS CodeVita Season 13**, demonstrating exceptional **problem solving**, **DSA**, and algorithmic complexity analysis skills under strict time constraints.
- Secured **First Prize** in **Inter-Department Major Project Presentation**, articulating architecture, machine learning challenges, and AI solutions to a panel of judges.
- Awarded **First Runner-Up** in **Sprint Hackathon**, showcasing the ability to handle ambiguous technical challenges and rapidly deliver functional AI prototypes.